

forums are typically information-seeking in nature, the ugly spectre of misbehaviour raises its head often on these platforms, with individuals posting questions intended to troll specific groups/communities, to incite hate speech, or have objectionable content. Individuals also post opinions in the guise of questions, thereby inciting arguments on such CQA forums. Often rhetorical and insincere questions get posted by individuals, as a joke, on these forums. Hence content moderators need to filter such questions so that the CQA site contains good quality content that is not offensive in nature. Questions that are not true requests for information are termed as insincere. Quora, in fact, provides guidelines on what constitutes insincere questions and has mandated that these get filtered; individuals who frequently post them can be banned from the site.

As mentioned before, automated approaches are needed, in addition to manual moderation, to filter insincere questions on CQA sites. This is the task for insincere question classification. Quora has released a labelled data set for this problem, which is available at <https://www.kaggle.com/c/quora-insincere-questions-classification/data>. In fact, there is a Kaggle competition currently running for this data set. Hence it would be good for our readers to join the competition and develop hands-on techniques for addressing this challenge. Towards that end, let us dig further into this problem.

Given a question, the task is to either label it as an insincere question or not. This is a binary classification problem. Given that questions are typically short, this can also be considered a short text classification problem. Before we design a solution for this task, let us first try to understand what exactly can lead to a question being labelled as insincere. There are a number of reasons for this, which include the following.

- **Rhetorical questions:** These are non-neutral and convey an opinion or take a stand. For instance, a true request for information is not about taking a stance or expressing an opinion. On the other hand, questions that try to frame an opinion as a question are not requests for information and are typically insincere. Such questions are rhetorical in nature. A few examples of such questions are given below.
 - Q1: *Can we all now admit that President Trump doesn't really want Congress to pass legislation replacing DACA to protect dreamers?*
 - Q2: *How much more political fumbling will it take for Republicans to turn on Trump?*

Question Q1 carries the implicit opinion that President Trump is against DACA and Q2 carries the implicit opinion that he has been fumbling in his role

as the US president. While Q1 and Q2 are phrased as questions, neither of them are information-seeking in nature, which is why we call them insincere and rhetorical.

- **Disparaging/inflammatory questions:** These are intended to insult or attack certain groups of people. For instance, sexist and trolling comments are often posted as questions that are intended to offend specific minority groups or individuals. A few examples of such questions are:

- Q3: *Do Trump haters, by definition, realise that they are bigots?*
- Q4: *When will Hillary Clinton finally pay for her crimes and go to jail?*
- Q5: *Why are there so many dirty Jews on Quora?*

On examining Q5, it is clear that the question does not seek any information and is intended to be disparaging and hurtful to a specific group of people, namely Jews. Similarly, Q4 is intended to disparage an individual namely Hilary Clinton.

- **Hypothetical or unreal questions:** These are not real and meant to be fictitious. Typically, they are hypothetical and have an unreal context. Hence they are not true requests for information. Examples of such questions include:

- Q6: *Have you ever shot your mom?*
- Q7: *If both Honey Singh and Justin Bieber fall from the fifth floor, who will survive?*

It is easy to see that both Q6 and Q7 portray unreal situations and are not intended to be requests for information.

- **Questions with objectionable content:** These use sexual content for shock value.

Questions belonging to any of the above categories are termed as insincere. A detailed description of insincere questions, with examples, can be found at <https://www.quora.com/What-is-an-insincere-question>.

Now that we have defined what constitutes an insincere question, let us look at how we can design a text classifier to solve this task. As we had discussed, we can start off with a simple Bag of Words classifier for this binary classification. We can try different classifiers such as Support Vector Machines, Random Forests and Gradient Boosted Decision Trees with Bag of Words as features.

We can augment the Bag of Words vector with additional features. For instance, we can capture Parts of Speech Tags information for the text, and the sentiment contained in the text as features. Certain dependency relations in the text can also serve as features. For instance, dependency relations such as the adjective modifier relation or an adverbial modifier relation can often serve to indicate opinions. We can use the presence of these dependency relations as an additional feature for an objectivity classifier. We can augment our